PrejSense: Non-intrusive Gaze-Informed Prediction of Implicit Prejudice Against Machine for Proactive Interface Adaptation

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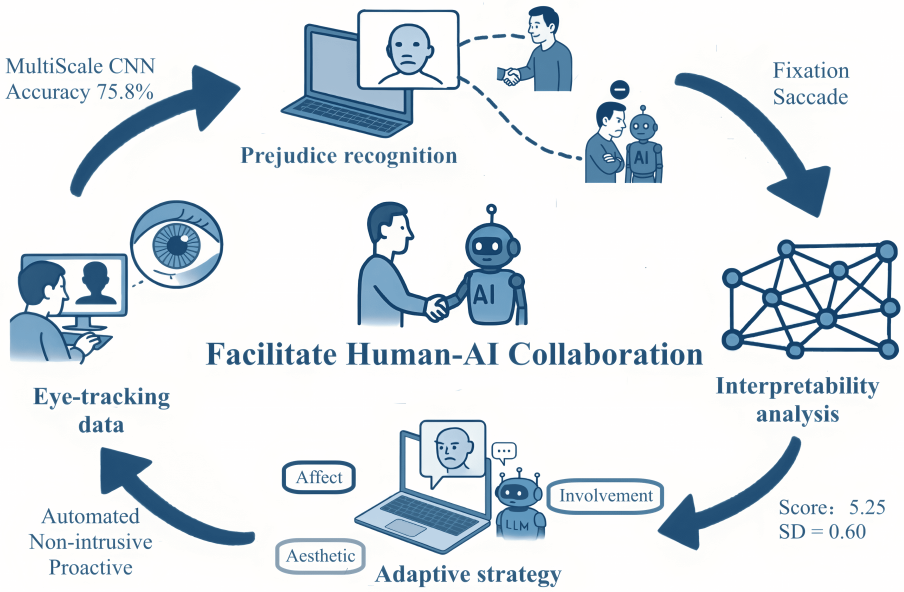


Figure 1: Overview of PrejSense in human-AI collaboration loop.

Abstract

Implicit prejudice against machine can critically undermine the effectiveness of human-AI collaboration by limiting trust, cooperation, and system acceptance. Despite its significance, such prejudice is rarely addressed in real world usage and is often assessed retrospectively through self-report measures or implicit association test, lacking automated and adaptive interface essential for responsive system design. Drawn from dual process theory, we propose and test PrejSense, a proof-of-concept adaptive interface loop that uses non-intrusive gaze signals to predict users' prejudice states and adapt interface support accordingly. First, we develop interpretable deep learning models to predict prejudice from gaze features. Second, we develop adaptive interface using LLM agents to dynamically respond to estimated prejudice states, with expert-informed refinement. Third, we build a full-stack system for automated prejudice prediction and adaptive interaction, enabling non-intrusive integration into collaborative AI systems. We conduct a user testing experiment to evaluate the effectiveness of PrejSense. Compared with baseline, PrejSense significantly enhanced emotional trust and showed marginal positive effects on cognitive trust and WOA. These results provide preliminary evidence for state-contingent prejudice-aware adaptation in a low-risk decision-support task. By combining gaze-informed prejudice prediction with adaptive interaction design, this work opens a design space for prejudice-aware adaptive interfaces while highlighting the need for larger-scale validation.

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CCS Concepts

• **Human-centered computing** → **User interface design**; **Eye-tracking**; • **Computing methodologies** → *Supervised learning*

by classification; *Machine learning algorithms*; • **Social and professional topics** → *User characteristics.*

Keywords

Implicit Prejudice against Machine, Adaptive Interaction Interface, Human AI Collaboration, Human-centered AI, Recognition System

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1 Introduction

Implicit prejudice against machine is an overlooked yet critical barrier to realizing the full value of artificial intelligence (AI). AI systems now inform decisions in healthcare [68], finance [124], and education [42], promising gains in efficiency and accuracy. However, recent evidence shows that human–AI interaction is frequently distorted by implicit prejudice that people bring to these collaborations [27, 63, 112]. First, humans with specific implicit prejudice may either over-rely on an AI's output or reject it outright, as a pattern captured by automation bias and algorithm aversion [31, 112]. Such polarized responses degrade joint decision quality, erode user confidence, and threaten the perceived fairness and transparency of outcomes [76, 83]. From cognitive psychology perspective, current research attributes reactions with implicit prejudice to the dual-process structure of human reasoning [38, 61, 75]. Faced with complex or time-pressured tasks, people default to fast, intuitive System 1 processing rather than the slower, analytical System 2, making them susceptible to heuristics and stereotypes about machine competence [61, 75]. Therefore, the dual-process perspective implies that technical improvements alone are insufficient. Designers should integrate psychological principles that counteract intuitive shortcuts and promote reflective engagement, thereby mitigating prejudice and unlocking the promised benefits of AI-supported decision making [16]. Addressing this challenge requires further exploration on how to predict and manage implicit prejudice against machine in human-AI collaboration.

Predicting humans' implicit prejudice is pivotal for fostering effective human–AI collaboration, yet it remains methodologically challenging. Most existing attempts still depend on explicit self-report instruments or standardized questionnaires, methods that intrude on natural interaction and are vulnerable to social-desirability bias [18, 112]. Behavioral tests such as semantic-priming paradigms [103] and the Implicit Association Test (IAT) [40, 45, 112] automate part of the assessment, but they likewise require overt user responses and cannot be seamlessly embedded in natural human–machine dialogues. More recently, language-based machine-learning models have begun to infer implicit prejudice directly from textual data [13]. Although such approaches remove the need for questionnaires, they depend on the availability of sufficient linguistic input[9, 130]. Physiological signals provide a promising alternative because they can unobtrusively reveal latent cognitive states. Preliminary studies leveraging electroencephalography (EEG) and electrodermal activity (EDA) have demonstrated the feasibility of predicting implicit prejudice without interrupting the user [14, 59]. Nonetheless, the reliability and generalisability of these biomarkers across tasks and user populations remain inconclusive, and systematic investigations of other easily accessible indicators are scarce. Eye-movement data, in particular, hold considerable potential. Gaze dynamics have been shown to encode fine-grained attentional and evaluative processes in creative search and human–AI interaction [49, 71, 90], yet they have not been systematically exploited for implicit prejudice detection. In addition, a further limitation of the current implicit prejudice detection is interpretability. Most detection pipelines rely on opaque "black-box" models that privilege predictive accuracy over explanatory insight, hindering theoretical understanding of psychological model and practical trust [3, 107].

An important design mandate is that implicit-prejudice against machine sensing should be automated and non-intrusive if it is to be deployed in real-world settings. Automated interfaces can capture physiological data streams in real time, thereby guaranteeing timeliness and personalization without additional user effort [109]. Equally important, non-intrusive sensing minimizes interruptions by operating in the background, preserving the user's primary task focus and reducing cognitive load [70, 114]. Gaze is especially relevant to this design mandate because eye contact and gaze behavior are core nonverbal signals for information seeking, interaction regulation, and social evaluation [64]. Prior prejudice research further shows that implicit attitudes can be expressed through nonverbal behaviors such as visual contact and blinking [34, 35]. Recent eye-tracking studies also suggest that implicit or attitudinal biases can be associated with gaze allocation and fixation patterns in social perception contexts [41, 111]. Wearable or contact-less devices such as the AirSpecs smart-glasses platform [130] demonstrate how continuous physiological monitoring can be folded seamlessly into everyday routines. Compared with EEG, EDA, or ECG, webcam-compatible gaze sensing is better aligned with real-time adaptive interfaces because it can be collected with lower equipment burden and without interrupting the primary task [23, 93]. Despite these advances, implicit-prejudice recognition remains largely overlooked. Existing automatic systems target comfort, workload, or personality traits rather than implicit prejudice against machines [49]. We therefore use gaze as a predictive behavioral correlate of prejudice-associated user states, rather than as causal evidence of implicit prejudice. This gap limits the deployment of day-to-day human–AI interactions while preserving user comfort and privacy.

Crucially, predicting prejudice is a means to an end. Once an AI system is aware of the user's implicit prejudice it can proactively and adaptively adjust its behavior to maintain collaboration quality. Proactive interfaces that act before explicit user requests, whether in conversational assistants [36, 116] or toxicity-aware keyboards [126], have been shown to enhance decision-making and engagement. Adaptive teamwork agents similarly improve joint performance by modulating their autonomy based on evolving task demands [51] or by tuning sensory complexity to match cognitive load [22]. Non-intrusive personalization grounded in physiological signals [49] demonstrates the feasibility of seamless, context-aware adaptation. Building on these insights, integration of implicit prejudice prediction into an interactive layer that can reformulate explanations, alter tonal style, or request clarifications when prejudice is sensed, following best practices in proactive moderation and

adaptive dialogue [79, 94, 132]. To realize an adaptive interface capable of responding effectively to detected prejudice, employing large language models (LLMs) is advantageous due to LLMs proven capabilities in context-aware adaptation, flexible dialogue generation, and nuanced response modulation [78, 102]. Additionally, to validate LLM-based adaptive interfaces from the user's perspective, iterative evaluations using expert focus-group studies are required. Expert focus-group studies could systematically assess usability, transparency, and perceived fairness, providing critical feedback for refining interface design [11, 32, 73]. Thus, this research integrates LLM-based adaptive interfaces informed by expert-informed formative refinement, thereby establishing a methodological foundation for implicit prejudice-aware loop in human-AI interactions.

Motivated by these challenges, our work proposes a novel user interface PrejSense, which dynamically predicts and adapts to the user's implicit prejudice against machine. PrejSense aims to address four key research questions at the prediction of implicit prejudice and adaptive AI interfaces. Specifically:

RQ1: How can eye-tracking features predict users' implicit prejudice against machine?

RQ2: What gaze features are associated with implicit prejudice model predictions?

RQ3: How can systems proactively and adaptively optimize user experience based on predicted implicit prejudice?

RQ4: How to build an automated, non-intrusive system that integrates gaze-based state prediction with prejudice-aware interface adaptation?

To answer these four RQs, we developed and tested PrejSense, a non-intrusive system for implicit prejudice recognition and proactive adaption. As shown in Figure 2, PrejSense consists of three modules. In module 1, we collected eye-tracking data from participants engaged in collaboration decision tasks, employing interpretable deep learning models to predict implicit prejudice based on extracted eye movement features. In module 2, we implemented a proactive, adaptive interaction interface using LLMs tailored specifically to users' implicit prejudice states identified by module 1. The effectiveness, transparency, and perceived fairness of this adaptive interface were validated and iteratively refined through expert focus-group interviews. Module 3 integrated module 1 and module 2 by developing an automated, real-time, and non-intrusive system, designed to integrate seamlessly into practical AI interfaces without disrupting user workflow. We conducted a user testing experiment to evaluate PrejSense. Results showed that PrejSense improved emotional trust and showed positive effects on cognitive trust and WOA in a low-risk task, supporting the feasibility of the prejudice-aware adaptive interface loop. Thus, PrejSense systematically addressed the estimation, interpretation, real-time integration, and adaptive use of prejudice-associated user states, forming a preliminary proof-of-concept approach to enhancing human-AI collaboration.

This work makes three contributions.

- We present an end-to-end framework that translates interpretable gaze features into actionable prejudice-aware interface adaptation. By combining gaze-based prediction, interpretability analysis, and LLM-mediated interaction design, the work moves beyond retrospective measurement of implicit prejudice and shows how a latent user state can be operationalized through adaptive changes in explanation, tone, and user control. We position this framework as a preliminary adaptive interface loop rather than a definitive psychological measurement tool.
- We develop a lightweight, deployable proof-of-concept system that integrates non-intrusive sensing, real-time prejudice prediction, and adaptive interaction into a practical pipeline, with the adaptive mechanism refined through expert focus groups and its effectiveness supported by small-scale user testing showing improved trust and reduced algorithm aversion.
- Rather than positioning the current model as a final solution, We establish an initial benchmark and a practical roadmap for future research on prejudice-aware adaptive interfaces by providing a reference pipeline, an empirical baseline, and concrete directions for optimizing prediction accuracy, multimodal modeling, deployment robustness, and privacy-aware adaptation in human-AI collaboration.

2 Related Work

2.1 Implicit Prejudice Against Machine

Implicit prejudice against machines has been empirically shown to undermine trust, acceptance, and the overall effectiveness of human-AI collaboration. Drawing on dual-process accounts, prior work suggests that users' responses to AI are often shaped by rapid, intuitive evaluations rather than fully deliberative reasoning [38, 75]. For example, implicit negative associations toward AI can hinder trust and adoption by creating psychological barriers to relying on algorithmic support and by contributing to algorithm aversion, leading users to discount potentially beneficial AI advice [27, 63, 112]. Beyond trust and adoption, such prejudice may also discourage consumer acceptance of AI-enabled services and shape interaction quality in highly automated settings such as commercial AI applications and vehicles [129]. Together, these findings suggest that implicit prejudice against machines is an important but still underexplored barrier to effective human-AI collaboration.

Despite growing evidence of these negative effects, two important gaps remain. First, while prior studies on implicit prejudice and dual-process theory provide useful theoretical background [7, 38, 75, 118], it remains unclear how implicit prejudice against machines is reflected in observable interaction processes during human-AI collaboration. Existing work has mainly examined downstream outcomes such as trust, aversion, or acceptance, offering limited insight into the behavioral or physiological patterns through which machine-directed prejudice may manifest [7, 118, 123]. For this reason, psychological theory in this work serves primarily as an interpretive lens for understanding prejudice-related interaction signals, rather than as a basis for making strong causal claims about underlying mechanisms. Second, practical approaches for recognizing and managing implicit prejudice against machines remain scarce. Although prior research has documented its consequences for trust, consumer acceptance, and system effectiveness [27, 112, 129], there are still few methods for systematically estimating and responding to prejudice in realistic interaction settings.

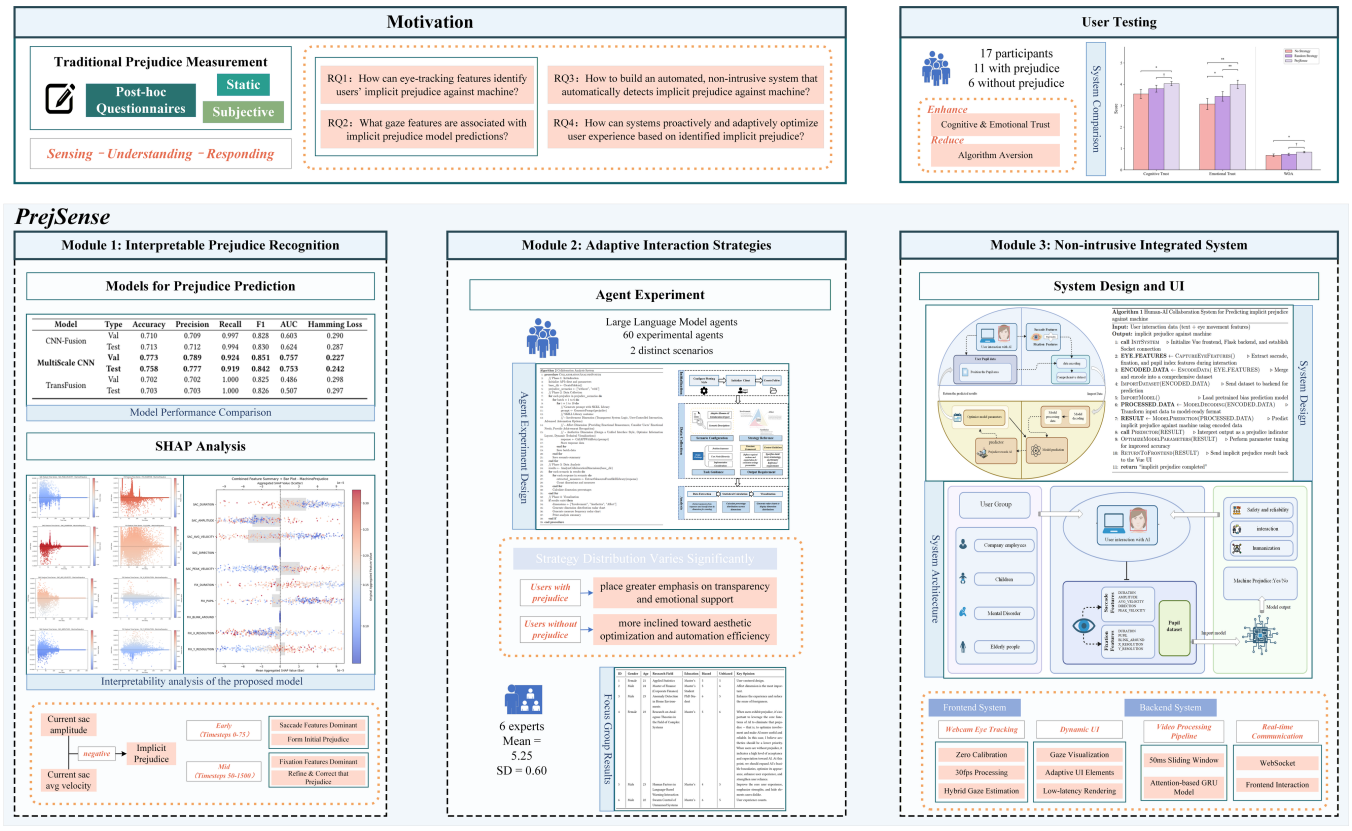


Figure 2: Research Framework

This gap motivates the development of approaches that can identify prejudice-related patterns during interaction and support more effective human-AI collaboration through adaptive interface design.

2.2 Measurements of Implicit Prejudice

Traditional measures of implicit prejudice primarily rely on structured assessments and standardized questionnaires [18]. Common examples include semantic priming tasks, which quantify prejudice toward robots through reaction-time differences to robot versus human primes [103], and the Implicit Association Test (IAT), which captures automatic evaluative associations that are less susceptible to conscious control or self-presentation bias [40, 45, 112]. More recently, computational approaches have extended these measurements beyond direct testing. For example, Bhatia and Walasek predicted IAT scores from linguistic data using word embeddings [13]. Together, these methods reflect a shift from explicit assessments toward computationally enhanced inference of implicit prejudice.

However, questionnaire-based and language-based methods have clear limitations in interactive settings. Self-report assessments interrupt ongoing human-AI interaction and may affect task performance [9, 130], while linguistic approaches may not fully capture subtle implicit processes in complex contexts [13]. These limitations have motivated recent work on physiological signals as alternative indicators of implicit prejudice [14, 59]. Early studies suggest that signals such as EEG and EDA may support implicit prejudice detection, but current evidence remains preliminary and their reliability in practical settings is still unclear [14, 59].

This literature points to several opportunities for PrejSense. First, physiological signals may enable more unobtrusive recognition of implicit prejudice during natural interaction. Eye movements are especially promising because they can reflect subtle cognitive responses without interrupting the user [49, 71, 90]. This choice is also supported by classic and recent evidence that gaze can reflect attention, regulation, social evaluation, and attitudinal orientation in social contexts [35, 41, 64, 111]. Yet compared with EEG and EDA, eye-tracking remains underexplored for this purpose. Second, existing prediction approaches often emphasize accuracy over interpretability, limiting understanding of which signals are associated with prejudice-related predictions [3, 14, 59, 107]. Third, implicit prejudice specifically directed toward machines remains less studied than prejudice toward human social groups [7, 118]. Motivated by these gaps, our work focuses on predicting implicit prejudice against machines from physiological signals, with particular emphasis on eye movements as unobtrusive indicators, while using interpretability analysis to provide theory-informed insight into model predictions rather than causal proof of psychological mechanisms.

2.3 Automated and Non-intrusive Recognition

Automated and non-intrusive systems have attracted increasing attention for their ability to support efficient, personalized, and minimally disruptive interactions. Automated systems can respond to user states by processing physiological or behavioral signals in real time, while non-intrusive systems operate in the background to reduce interruptions and cognitive burden [70, 109]. Prior work has demonstrated these benefits across diverse settings, including wearable sensing for comfort and cognitive-state monitoring [130], contactless assessment for adaptive interaction [58], and workflow-integrated AI assistance in domains such as programming and healthcare [110]. Together, these studies show that automated and non-intrusive systems can enhance user experience through context-aware and seamless support. Recent reviews further show that eye tracking has become an important HCI method for interaction design, cognitive-state assessment, gaze estimation, and adaptive interface research [67].

However, their application to implicit prejudice against machines remains largely unexplored. Existing work shows that such systems can support human-AI interaction by inferring user states and adapting responses in real time [44, 49], while reducing disruption during ongoing tasks [70, 114]. Yet current systems mainly target cognitive states, comfort, or productivity rather than prejudice recognition, especially implicit prejudice against machines. This gap motivates our work to develop an automated and non-intrusive system for predicting machine-directed implicit prejudice, with the goal of supporting more effective and trustworthy human-AI collaboration.

2.4 Proactive Adaptive Interaction

Adaptive proaction interfaces have advanced personalized and timely user support by adjusting system behavior based on user context [43, 116]. Prior work has demonstrated these benefits across multiple settings. In conversational interaction, proactive agents can provide timely feedback without explicit user input, thereby shaping decision-making and reflection [36]. In task-oriented environments, adaptive systems can dynamically adjust autonomy or interface complexity in response to changing user and task states, improving performance and engagement [22, 51]. Non-intrusive personalization methods further show that physiological and behavioral signals can support context-sensitive adaptation without interrupting the user [49]. Together, these studies suggest that proactive and adaptive interaction can effectively enhance user experience through context-aware and personalized support.

However, systems that adapt specifically to implicit prejudice against machines remain largely unexplored. Existing work has mainly adapted to explicit behaviors, task context, or general physiological states, with limited attention to machine-directed implicit prejudice. This gap is especially important because such prejudice may shape users' trust, acceptance, and cooperation in human-AI interaction. Motivated by this gap, our work develops an LLM-based adaptive interface that responds to predicted implicit prejudice against machines [79, 94, 132]. We further incorporate expert focus group evaluation to refine the adaptive mechanism and assess its practical usability, transparency, and relevance in real-world scenarios [11, 32, 73]. This design extends proactive and adaptive interaction toward prejudice-aware interface adaptation in human-AI collaboration.

3 PrejSense

3.1 Module 1: Interpretable Prejudice Recognition

3.1.1 *Data and Method.*

Research Data. Module 1 collected eye-tracking data from 40 participants during an interactive experiment. The protocol comprised two phases (see Appendix Figure 1). In Phase 1, we measured participants' implicit prejudice using the Implicit Association Test (IAT). Implicit prejudice refers to social preferences outside conscious control [72, 112]. We followed Greenwald et al.'s method of computing the mean response-time difference between compatible and incompatible tasks to classify participants as implicitly prejudiced or non-prejudiced [1, 46, 112]. The measurement details are shown in appendix B. In Phase 2, participants performed 50 decision-making trials with the assistance of the AI, which provided both recommendations and explanatory reasoning for each task. Participants reviewed these outputs before making their own judgments. Eye-movement recordings were then split into fixation-related and saccade-related features (see Appendix Table 1). Finally, we constructed a dataset using eye-movement features as input features and implicit prejudice of participants as the output feature. To prepare the eye-tracking data for model training, we applied a standard preprocessing pipeline. Full preprocessing details are provided in Appendix E.

Model Training. We implemented three deep learning frameworks (CNN-Fusion, MultiScale CNN, and TransFusion) to predict implicit prejudice from eye-tracking data. A Linear model was established as the baseline for performance comparison. All models' design details are summarized in Appendix Table 2. The eye-tracking signals were preprocessed. We then split the dataset into training, validation and test sets (80:10:10) via stratified sampling. To address class imbalance, an inverse class frequency weighting strategy was adopted, establishing weighted loss function mechanisms to optimize model learning. Each model was trained using the AdamW optimizer alongside the Weighted BCEWithLogitsLoss function [66]. To ensure robustness, a five-fold stratified cross-validation scheme was applied, incorporating dynamic learning-rate scheduling and early stopping to identify the best-performing model based on test accuracy. Finally, performance was evaluated using multiple metrics, including accuracy and F1 score.

To interpret the best-performing model for each prediction target, we employ SHAP, a feature attribution method grounded in Shapley values [81]. Detailed implementation procedures are provided in Appendix H.

3.1.2 *Results and Discussion.*

Module 1 employs three deep learning frameworks to construct an implicit prejudice prediction model utilizing eye-tracking behavior and interactive contextual data. For brevity, Table 1 reports only the key test-set metrics (Accuracy, F1, and AUC). The full validation and test results are provided in Appendix Table 6. Additional model-validity analyses are reported

Table 1: Model performance comparison on the test set. Full validation and test metrics are reported in Appendix Table 6.

Model	Accuracy	F1	AUC
Linear (baseline)	0.703	0.826	0.505
CNN-Fusion	0.713	0.830	0.624
MultiScale CNN	**0.758**	**0.842**	**0.753**
TransFusion	0.703	0.826	0.507

in Appendix G, including fold-level 95% confidence intervals, the confusion matrix of the best-performing MultiScale CNN fold, and false-positive and false-negative patterns. The experimental results suggest that eye-tracking data, as a form of nonverbal behavior, holds potential for assessing implicit prejudice [92]. Among the evaluated architectures, the MultiScale CNN model demonstrates superior overall performance. Its test set accuracy (75.8%), precision (77.7%), F1 score (0.842), and AUC (0.753) significantly outperform both the Linear baseline (Accuracy: 70.3%, AUC: 0.505) and the other deep learning variants, specifically CNN-Fusion (Accuracy: 71.3%, AUC: 0.624) and TransFusion (Accuracy: 70.3%, AUC: 0.507). Furthermore, MultiScale CNN achieves the lowest Hamming Loss (0.242). This confirms that the multi-scale transformation in the MultiScale CNN model can enhance feature representation and improve classification accuracy, while its feature fusion capability effectively captures eye-movement patterns related to implicit prejudice.

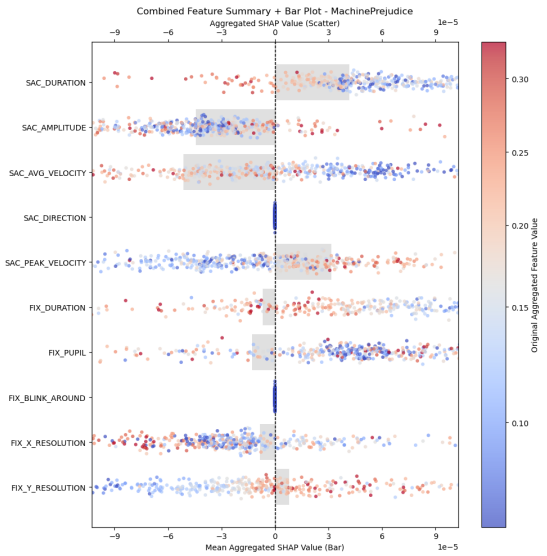


Figure 3: SHAP summary and bar plots.

Note. SHAP values indicate model-based feature attributions and should not be interpreted as causal evidence of psychological mechanisms.

To further examine which eye-movement features were most relevant to the model's predictions, we applied SHAP to the best-performing MultiScale CNN model. Rather than serving as direct evidence of psychological mechanisms, SHAP provides a model-based attribution view that helps relate prediction patterns to prior

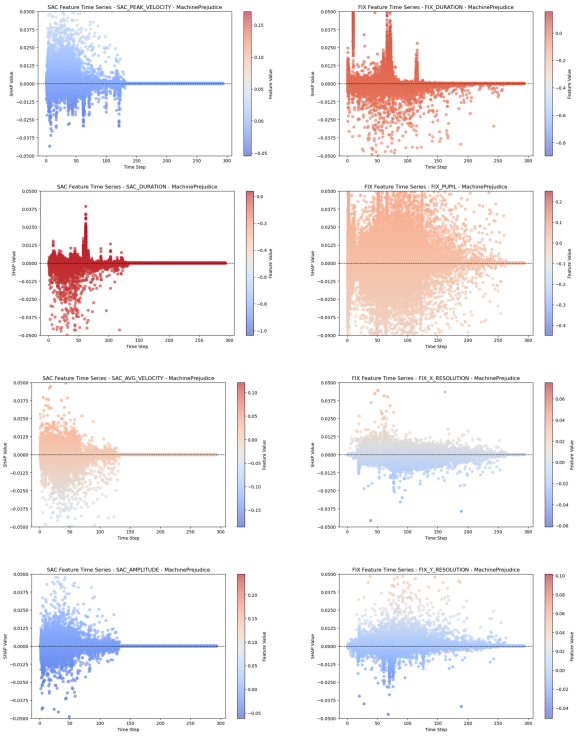


Figure 4: Distribution of SHAP values across timesteps.

Note. Temporal SHAP patterns describe model prediction behavior rather than causal psychological processes.

cognitive accounts [84]. Figure 3 shows the overall contribution of each eye-movement feature, and Figure 4 illustrates how these contributions vary over time.

Two patterns are noteworthy. First, saccade-related features appeared particularly influential. In Figure 3, *current sac amplitude* and *current sac avg velocity* showed relatively strong contributions and were negatively associated with predicted prejudice. In particular, higher saccade velocity was more often observed in samples predicted as low prejudice. This pattern is broadly consistent with prior eye-tracking work suggesting that faster scanning can accompany more fluent processing, whereas slower gaze behavior may reflect greater hesitation or deeper inspection [80, 106]. We therefore interpret these features as informative behavioral markers of prejudice-related differences, rather than as definitive evidence of an underlying cognitive mechanism.

Second, the temporal SHAP analysis suggests that different feature types may matter at different stages of interaction. As shown in Figure 4, saccade-related contributions were concentrated earlier in the sequence, whereas fixation-related contributions became more visible in the middle phase. This pattern suggests that early gaze behavior may capture rapid orienting, while later fixations may reflect more sustained processing of task-relevant information [98]. Such a temporal structure is compatible with dual-process accounts of fast versus more deliberate processing [61, 104], although our results should be understood as correlational and predictive rather than mechanistic.

3.2 Module 2: Proactive Adaptive Interaction

3.2.1 *Research Method.*

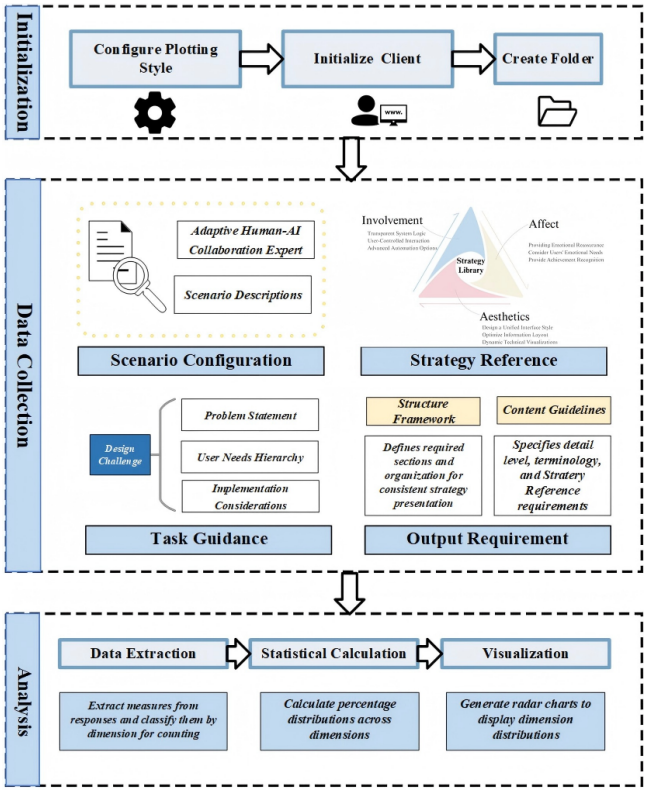


Figure 5: Agent Experiment Design

LLM-based Experiment. Module 2 employs an LLM-based experimental framework, using Deepseek-R1 as the testbed [28] to investigate how to develop adaptive interaction interfaces based on predicted prejudice results. Specifically, we conducted 120 agent-based experiments across two scenarios—one in which users exhibited implicit prejudice against the system and one in which they did not—with the aim of enhancing user experience by tailoring responses to individual user characteristics. Interaction strategies were organized along three dimensions—Involvement, Aesthetics, and Affect [113]. In recent years, personalized design and deployment in human-AI collaboration contexts have received widespread attention [17, 24], making user-specific strategy adaptation a key design consideration. For analysis, we calculated and compared the distribution of strategy dimensions selected by the LLM across 60 experiments in each scenario and extracted the most frequently applied measures. The detailed workflow is depicted in Figure 5. To comprehensively analyze how different foundational models generate adaptive strategies, module 2 employs a comparative LLM-based experimental framework. The details of comparison method and results as shown in Appendix I.

Expert Focus Group Interview. To assess the professional rigor of the LLM's strategy selection, focus group interviews were conducted [2, 11, 32, 73]. Participants were recruited via purposive sampling to ensure appropriate backgrounds and domain expertise. Eligible participants were required to have at least one of the following forms of relevant expertise: publications in HCI, human-AI interaction, adaptive interfaces, or trust in automation; research or industry experience in AI-assisted decision-making or UX evaluation; or practical experience designing or evaluating LLM-based interactive systems. The semi-structured interviews explored the categorization of interaction strategies in the LLM agent experiments, the reasonableness of strategy adjustments across different contexts, the impact of dynamic implementation on user experience, and the practical applicability and limitations of the findings [56]. Sessions were held in a quiet, distraction-free environment to encourage open and thoughtful participation. After the discussion, participants rated the adaption interface's professionalism on a seven-point Likert scale, providing quantitative feedback on its perceived rigor and credibility.

3.2.2 *Results and Discussion.*

Module 2 examines LLM agents as adapt interaction interface. Appendix K Figure 2 illustrates the dimensional proportions of different models with(a) and without(b) prejudice. Appendix L Figure 3 highlights the most frequent measures in each case. Two patterns are notable. First, under prejudice, the generated strategies prioritized transparency and user control. Involvement was the dominant dimension (68.3%), with common measures including Transparent System Logic and User-Controlled Interaction. This suggests that when prejudice is present, users may benefit from interfaces that make system reasoning more visible and allow greater control over algorithmic assistance, which is broadly consistent with prior work on transparency, predictability, and trust in automated systems [53, 74]. Affect also remained important (27.8%), suggesting that brief supportive feedback may help reduce uncertainty and resistance [97]. From a technology-acceptance perspective, these strategies may also improve perceived usefulness and ease of use [26, 119, 120].

Second, under non-prejudice, the strategies focused more strongly on presentation efficiency and interface fluency. Aesthetics accounted for 73.9% of selected strategies, primarily through Optimize Information Layout and Dynamic Technical Visualizations. This suggests that when users are already receptive to AI suggestions, adaptation can shift from trust repair toward efficiency enhancement through clearer layout and reduced interaction effort. Such a pattern is consistent with prior research showing that simplified presentation can increase cognitive fluency [5] and that improved interface design can enhance usability and operational efficiency [33]. It is also in line with work suggesting that users with lower resistance to algorithmic support may place greater value on smooth interaction and efficient assistance [19].

To obtain expert-informed formative feedback on the LLM-generated strategies, we convened a focus group of six human–AI collaboration experts who rated strategy quality on a seven-point Likert scale. The mean professionalism score was 5.25 (SD = 0.60), indicating above-average expert approval. Experts also offered tailored recommendations for the scenarios with prejudice and without prejudice. In prejudice-sensitive contexts, they advised leveraging historical data to build detailed user profiles and providing periodic reports on AI outcomes to bolster system trust. For users without prejudice, they recommended integrating cross-domain features,

designing long-term engagement mechanisms, and using precise demand forecasting to enhance experience. Additionally, they advocated granting users without prejudice greater control to achieve more efficient and personalized interactions. Module 2 highlights the potential of LLM agents to optimize user interactions based on predicted prejudice, offering practical guidance for designing more personalized and effective strategies.

3.3 Module 3: Non-intrusive Integration System

3.3.1 System Design and UI. We propose a prototype of real-time prejudice prediction and adaption system based on module 1 and module 2. System interaction structure as shown in Appendix Figure 4, the front end using VUE 3 development in the user face dynamic change information, back end using lightweight architecture Flask data processing, import model for analysis prediction, and adaption. Between Socket programming interface network communication. Lightweight development makes PrejSense easily applied to the existing human-AI collaboration system.

3.3.2 System Architecture. As shown in Appendix N Figure 5, Module 3 operationalizes PrejSense as a real-time and non-intrusive pipeline that connects the prejudice-recognition model in Module 1 with the adaptive interface in Module 2. In deployment, the front end embeds WebGazer to continuously capture webcam-based gaze coordinates and timestamps with low overhead [57, 71]. These raw signals are then cleaned and segmented into fixation and saccade events, from which the eye-movement features required by Module 1 are computed [99]. The resulting feature sequences are sent to the pretrained prejudice prediction model, and the predicted state is passed to Module 2 to trigger prejudice-aware interface adaptation in real time. In this design, WebGazer serves as the online sensing layer, while prejudice prediction and adaptive response are handled by Modules 1 and 2, respectively. Detailed implementation details are provided in the Appendix O.

3.3.3 Real-time Performance. To examine whether PrejSense can operate at interactive speeds in deployment, we benchmarked the latency of the PrejSense. Following prior work on webcam-based gaze sensing and online eye-movement processing [37, 57], we measured the elapsed time from browser-side webcam frame acquisition to the completion of the prejudice estimate and the corresponding interface update. On a standard laptop with a built-in webcam, the system achieved a mean end-to-end latency of 68.7 ms (SD = 6.1 ms; range: 59.4–81.9 ms) across ten repeated 60-second trials, while sustaining an effective throughput of 17.2 fps (SD = 1.4 fps; range: 15.0–19.6 fps). This result is slightly below that of lighter gaze-only pipelines, which is expected because PrejSense includes additional stages beyond gaze capture. Even so, the measured runtime remained stable and fast enough to support responsive prejudice-aware adaptation in practical human-AI collaboration.

3.3.4 Ethics and Privacy Concerns. Because PrejSense uses webcam-based gaze and pupil-related signals to infer users' implicit prejudice and adapt the interface in real time, it raises important privacy and ethical concerns. Prior work has shown that eye-tracking data can support biometric identification and may reveal sensitive traits, internal states, and preferences beyond the immediate interaction context [25, 62, 69, 105]. To mitigate these risks, we designed the system around informed consent, data minimization, and bounded use. Before sensing begins, users are informed about what signals are collected, how they are processed, and how predictions may influence the interface. During use, webcam input is processed only to extract the fixation-, saccade-, and pupil-related features required for real-time adaptation, while raw frames are not stored or reused for secondary purposes. In addition, prejudice predictions are treated as probabilistic and context-dependent signals rather than stable personal attributes, and should only be used to support low-risk interface adaptations, such as providing more explanation, transparency, or user control. More broadly, prior HCI research suggests that the acceptability of camera-based sensing also depends on social transparency and privacy mediation in shared settings [12, 29]. Accordingly, users should be able to pause sensing or opt out of adaptive behavior at any time.

4 User Testing

4.1 Research Method

We designed and implemented a within-subject user testing experiment to evaluate the effectiveness of PrejSense, which dynamically adapts to users' implicit prejudice states. A total of 17 participants were recruited for the experiment (7 females and 10 males), including 11 users identified with implicit prejudice and 6 without. This study complied with the tenets of the Declaration of Helsinki and was conducted with the approval of the institutional review board at the authors' university. Informed consent was obtained from each participant. Three interaction scenarios were tested: (1) *No Strategy*, where the system only presented the AI's estimated result without any additional information; (2) *Random Strategy*, where the system randomly provided reasoning explanations or emotional text, without prejudice-aware adaptation; (3) *PrejSense*, which dynamically adapted support based on predicted users' implicit prejudice states. For users recognized with prejudice by PrejSense, the system added reasoning justifications and encouraging messages when displaying AI estimations, and provided emotional comfort after the task. For users without recognized prejudice, the interface emphasized key information and offered a "one-click adopt AI suggestion" function to enhance decision efficiency. Random Strategy used the same pool, amount, and presentation timing of interface support as PrejSense, including reasoning explanations, affective support, information highlighting, and adoption-oriented functions. The only difference was that Random Strategy assigned these supports randomly, whereas PrejSense matched them to the predicted prejudice state. This design allowed us to control for the mere presence of additional interface support and better isolate the value of state-contingent prejudice-aware matching. Each participant experienced all three interaction scenarios. The order of the three scenarios was counterbalanced using a Latin-square design, and the order of tasks within each scenario was randomized to reduce potential learning, fatigue, and carryover effects.

The experimental procedure followed Turel & Kalhan [112] and consisted of two phases (see Figure 6): (1) Weight Estimation Task – each participant completed three rounds of image-based weight estimation, each associated with one of the interaction scenarios. In each round, participants were asked to estimate the weight of four individuals (two male, two female, varying in body type) using

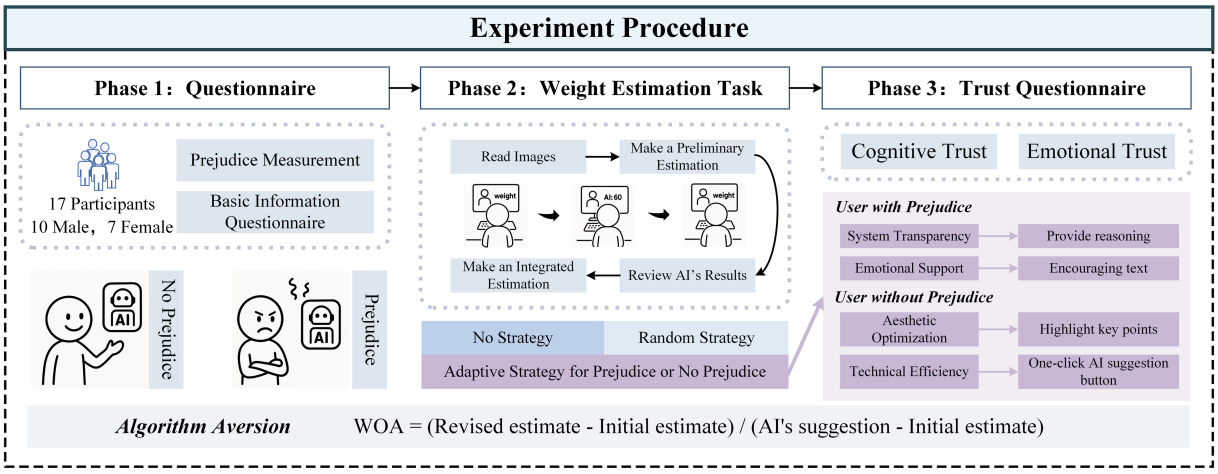


Figure 6: Experiment Procedure of User Testing

images sourced from the Jiang & Guo [60] database, which includes ground-truth weight labels. Participants first provided an initial estimate, after which the system presented an AI-generated estimation (constructed by adding 1–5% random noise to the ground-truth weight) [112]. All participants viewed identical AI values and experienced a delay to simulate computation time. Participants then decided whether and how to revise their initial judgment. (2) Trust Questionnaire – after completing each round, participants filled out a questionnaire assessing their trust in the AI system, covering both cognitive and emotional trust dimensions (each comprising five measurement items, adapted from Chen et al. [21], see Appendix Table 8). Additionally, following Turel & Kalhan [112], this study used WOA (Weight on Advice) to quantify algorithm aversion:

$$WOA = \frac{Revised\ estimate - Initial\ estimate}{AI's\ suggestion - Initial\ estimate}$$

and WOA values were truncated to fall within the range of 0 to 1, where a WOA of 0% indicates complete aversion to the AI's advice, and a WOA of 100% indicates full acceptance. To avoid pseudo-replication, the four trial-level WOA values within each scenario were first aggregated into a participant-level mean. Cognitive trust and emotional trust were also analyzed as participant-level scenario means. Therefore, the statistical unit of analysis was the participant-level scenario mean rather than the individual trial.

Because each participant experienced all three scenarios, we first analyzed the data using one-way repeated-measures ANOVA to test the overall within-subject effect of scenario. We report Greenhouse-Geisser corrected p values and partial eta squared (η_p^2) for the omnibus tests. Given the modest sample size, we also conducted exact sign-permutation tests for this planned contrast as an assumption-light robustness check. Full Holm-corrected pairwise comparisons across all three scenarios are reported in Appendix Table 9.

4.2 Results and Discussion

The experimental results demonstrate how our proposed system influenced users' trust levels and reliance on AI advice in a low-risk weight-estimation task. Figure 7 presents the comparison of

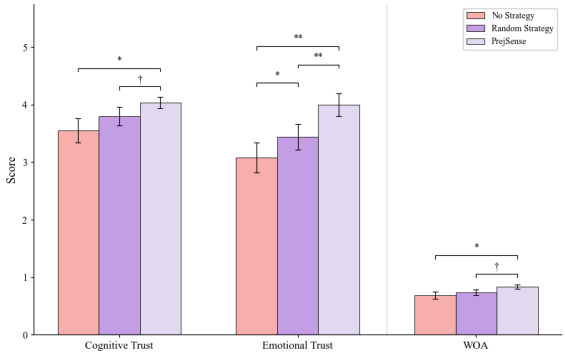


Figure 7: Comparison of participant-level user ratings and WOA across scenarios.

Note. Significance markers indicate two-sided paired-sample t tests for participant-level pairwise contrasts. †$p < .10$, * $p < .05$, ** $p < .01$.

user ratings across three different scenarios, while Tables 2 and 3, together with Appendix Table 9, provide the statistical results. The analysis yielded two key findings. First, PrejSense tailored to users' implicit prejudice results effectively enhanced trust in the interactive scenarios. The repeated-measures ANOVA showed significant main effects of scenario on both cognitive trust and emotional trust. In terms of cognitive trust, PrejSense resulted in higher user ratings (M = 4.04) compared to the no strategy condition (M = 3.55), with a statistically significant difference in the paired contrast ($p = .018$). Compared with Random Strategy, PrejSense showed a marginal positive effect on cognitive trust. The improvement in emotional trust was even more pronounced, with PrejSense showing significant advantages over both the no strategy and random strategy conditions across the paired tests and permutation checks. Second, PrejSense increased users' WOA, suggesting greater reliance on useful AI advice in this task. The repeated-measures ANOVA also showed a significant main effect of scenario on WOA. The average WOA score under the adaptive condition was 0.842, significantly higher

Table 2: Repeated-measures ANOVA results ($N = 17$).

Indicator	No Strategy	Random Strategy	PrejSense	$F(2, 32)$	p	p_{GG}	η_p^2
Cognitive trust	3.55 (0.86)	3.80 (0.67)	4.04 (0.41)	4.32	.022*	.029*	.213
Emotional trust	3.08 (1.07)	3.44 (0.91)	4.00 (0.81)	11.39	<.001**	.002**	.416
WOA	0.690 (0.249)	0.738 (0.204)	0.842 (0.152)	4.17	.024*	.027*	.207

Note. Values in the three scenario columns are means with standard deviations in parentheses. p_{GG} denotes Greenhouse-Geisser corrected p values. η_p^2 denotes partial eta squared. †$p < .10$, * $p < .05$, ** $p < .01$.

Table 3: Within-participant pairwise contrasts and exact sign-permutation checks.

Indicator	Comparison	Δ	90% CI	d_z	p_t	p_{perm}
Cognitive trust	Random Strategy – No Strategy	0.247	[-0.062, 0.556]	0.34	.182	.212
	PrejSense – Random Strategy	0.235	[0.016, 0.454]	0.46	.078†	.096†
	PrejSense – No Strategy	0.482	[0.161, 0.803]	0.64	.018*	.018*
Emotional trust	Random Strategy – No Strategy	0.353	[0.094, 0.612]	0.58	.030*	.037*
	PrejSense – Random Strategy	0.565	[0.284, 0.846]	0.85	.003**	.002**
	PrejSense – No Strategy	0.918	[0.473, 1.363]	0.87	.002**	.002**
WOA	Random Strategy – No Strategy	0.048	[-0.036, 0.132]	0.24	.334	.338
	PrejSense – Random Strategy	0.104	[0.013, 0.195]	0.48	.064†	.065†
	PrejSense – No Strategy	0.152	[0.047, 0.257]	0.61	.023*	.016*

Note. Δ is calculated as the latter scenario minus the former scenario. d_z denotes the standardized mean difference for paired comparisons. p_t reports the two-sided paired-sample t test. p_{perm} reports the two-sided exact sign-permutation test. Full Holm-corrected pairwise comparisons are reported in Appendix Table 9. †$p < .10$, * $p < .05$, ** $p < .01$.

than that under the no strategy condition and marginally higher than that under the random strategy condition, indicating a greater likelihood of users adopting AI suggestions. PrejSense not only improves users' cognitive and emotional trust, but also provides preliminary evidence of increasing reliance on AI advice compared with the baseline scenarios. Given the modest sample size, we interpret these findings as early-stage system evidence rather than definitive validation. Full Holm-corrected pairwise comparisons are reported in Appendix Table 9.

5 General Discussion & Conclusion

Figure 8 summarizes the full pipeline, from gaze-based prejudice predicting to prejudice-aware interface adaptation. This work explores how non-intrusive gaze signals can be used as predictive correlates of prejudice and translated into real-time interface support. More importantly, PrejSense translates interpretable gaze features into actionable prejudice-aware interface adaptation, turning a hidden user state into concrete changes in explanation, tone, and control. In this sense, the key contribution is not only prejudice prediction, but making it usable for adaptive interface design.

5.1 Theoretical Implication

This work offers two core theoretical implications for research on implicit attitudes, human-AI interaction, and adaptive systems. First, it suggests that fine-grained eye-movement dynamics can serve as interpretable behavioral correlates of machine-directed implicit prejudice. By showing that saccade-related features were informative for prediction, and that their temporal contributions were broadly consistent with fast-versus-deliberate processing accounts, this work extends the study of implicit attitudes from human social targets to AI systems [38, 61, 75]. Rather than establishing a direct psychological mechanism, the results provide a theory-informed and non-intrusive way to operationalize implicit prejudice during ongoing interaction. In this sense, eye-tracking complements traditional self-report and reaction-time measures such as the IAT by offering a behavioral lens on unconscious attitudes [7, 112].

Second, this work contributes to theories of proactive and adaptive systems by positioning implicit prejudice as a latent user-state variable for real-time interface adaptation. Prior research has mainly framed context-aware personalization around explicit input, task context, or coarse behavioral signals [36, 51, 121]. Our results suggest that implicit prejudice, inferred from physiological and behavioral traces, can function as a deeper form of context that drives automated interface adaptation [39, 55, 88]. In this way, the work extends adaptive-system thinking toward implicit user states [112]. PrejSense operationalizes a complete pipeline from sensing, to prediction, to prejudice-aware changes, thereby showing how a hidden state can be made actionable in interface design. More broadly, the combination of gaze-based sensing, LLM-mediated adaptation, and expert-informed strategy refinement provides a conceptual template for integrating implicit user states into adaptive interaction loops [48, 78, 82, 102]. From this perspective, the theoretical contribution is not only that implicit prejudice can be estimated as an interaction-relevant user state, but that it can be translated into a usable contextual resource for improving trust and collaboration in human-AI interaction [15, 94].

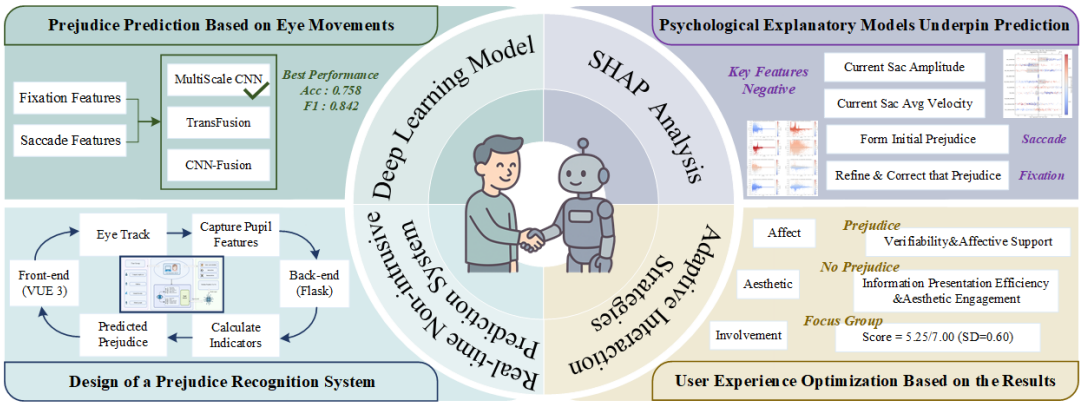


Figure 8: Research Results Summary

5.2 Practical Implication

The findings of this study offer two main practical implications for designers and developers of AI applications. First, PrejSense suggests that prejudice-associated state prediction can be incorporated into existing AI systems as a lightweight and deployable sensing-and-adaptation layer. Module 1 shows that eye-tracking features can support machine-directed implicit prejudice prediction, while Module 3 demonstrates how this capability can be integrated into a real-time and non-intrusive pipeline [100]. Because the system relies on commodity sensing and modular front-end/back-end coordination, it can be attached to existing interfaces with relatively low integration burden. This makes prejudice-aware adaptation feasible in applied settings where user trust and cooperation are critical [9, 95]. More broadly, the results suggest that implicit prejudice can be operationalized as an actionable signal for interface personalization.

Second, this study provides actionable design guidance for LLM-based interfaces that adapt to users' implicit prejudice. Findings from Module 2 indicate that when prejudice is detected, users may benefit more from transparency, controllability, and brief affective support, whereas users without evident prejudice may respond better to streamlined layouts and efficiency-oriented presentation [36, 51]. Practically, this suggests three compact design strategies: provide selective explanations when resistance is likely, offer adjustable levels of detail and control, and use lightweight supportive feedback to reduce uncertainty without increasing interaction burden [48, 78, 82, 102]. Rather than treating all users identically, practitioners can use prejudice-aware adaptation to tailor explanation, tone, and interface complexity to the user state at hand. In this sense, the main practical value of PrejSense is not only that it estimates a hidden user state, but that it turns that state into a usable design resource for improving trust, reducing algorithm aversion, and supporting smoother human-AI collaboration [15, 94].

5.3 Directions for Future Studies

Several future directions stem from this work. First, future research can treat PrejSense as a benchmark for optimizing gaze-based prejudice-aware adaptation rather than a final solution. The current predictive model, developed using a small sample size and achieving 75.8% accuracy. The central contribution of this study is to demonstrate an automated pipeline that translates gaze features into actionable prejudice-aware adaptation. Accordingly, the current predictive model should be understood as a promising starting point rather than an endpoint. There is significant potential for improvement through larger datasets and enhanced models. Second, our user testing was conducted in a low-risk weight-estimation task in which AI estimates were designed to be close to the ground truth. Therefore, higher WOA indicates greater uptake of useful AI advice in this setting, but does not by itself establish calibrated trust. Future studies should examine whether prejudice-aware adaptation can improve calibrated trust across broader task scenarios and varying levels of AI reliability. In addition, while this study provides valuable insights into the correlational patterns of gaze features, future studies should aim to establish causal relationships. Controlled experimental manipulations could help clarify the psychological mechanisms underlying implicit prejudice against machines. Finally, although this research has taken steps to address privacy and ethical concerns, more work is needed to ensure that real-time prejudice prediction and adaptation are conducted with appropriate safeguards. Future research should explore topics to mitigate privacy risks and prevent misuse.

5.4 Conclusion

In conclusion, PrejSense demonstrates a proof-of-concept path for translating non-intrusive gaze-informed estimates of machine-directed prejudice-associated states into real-time interface adaptation. By combining interpretable gaze-based prediction with an LLM-driven interaction layer, the system turns subtle eye-movement features into actionable prejudice-aware changes in explanation, tone, and control. This is the core contribution of the work: not only prediction a hidden user state, but making it usable for adaptive interface design. Across recognition, system integration, and user testing, PrejSense demonstrates a practical path for translating gaze features into actionable prejudice-aware adaptive interfaces that can support trust and reliance on useful AI advice in a low-risk decision-support task.

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A Appendix: Experiment Procedure of Prediction Module (Module 1)

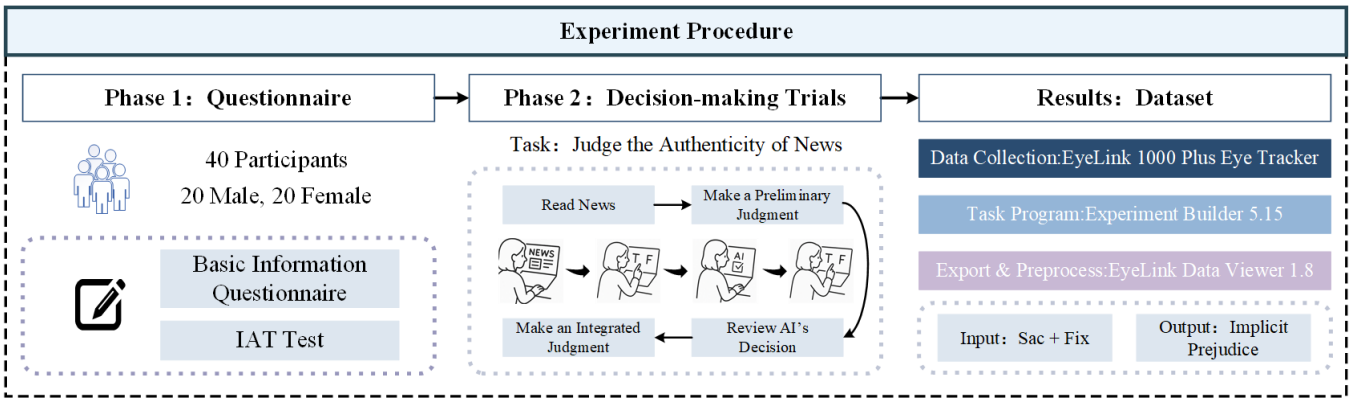


Figure 1: Experiment Procedure of Prediction Module (Module 1)

B Appendix: Construct Definition and Measurement of Implicit Prejudice Against Machine

Following prior work on *implicit prejudice against the machine*, we conceptualize this construct as an automatic, largely unconscious, and negatively valenced evaluative association toward AI or machine systems, especially in the domain of trustworthiness, rather than as a consciously reported attitude [112]. In this view, implicit prejudice reflects a latent associative bias that may shape how users respond to AI before they deliberately reflect on the system or accumulate diagnostic performance information [112].

Importantly, implicit prejudice against machine is *not* the same as algorithm aversion. The former is a latent psychological construct concerning automatic negative associations with AI, whereas the latter is a downstream behavioral tendency to discount, underweight, or reject algorithmic advice during decision making [112]. Thus, algorithm aversion is better understood as one possible behavioral consequence of implicit prejudice, rather than as the construct itself [112].

Our measurement strictly follows the IAT-based operationalization adopted in the attached papers. Before the main human-AI task, participants completed an Implicit Association Test (IAT) designed to capture automatic evaluative associations toward AI in the trustworthiness domain [112]. In the IAT, participants rapidly categorized stimuli belonging to a machine-related target category and a human reference category, together with evaluative attribute categories corresponding to *trustworthy* versus *untrustworthy* [112]. The logic of the measure is that response latency is shorter when two categories are more strongly associated at an automatic level. Accordingly, faster responding in blocks pairing AI with negative trust-related attributes, relative to blocks pairing AI with positive trust-related attributes, indicates stronger implicit prejudice against the machine [112].

Following the standard Greenwald-style scoring procedure used in the cited work, response times from compatible and incompatible IAT blocks were aggregated into an IAT D-score, i.e., a standardized difference in mean response latency across the relevant blocks [112]. More negative D-scores indicate stronger implicit prejudice against the machine, namely weaker automatic associations between AI and trustworthiness and/or stronger automatic associations between AI and untrustworthiness [112]. Conversely, less negative or positive D-scores indicate weaker prejudice or relatively more favorable implicit associations toward AI [112]. In line with the attached UIST paper, this IAT-derived indicator serves as the operational label of participants' implicit-prejudice state in subsequent analyses.

To further clarify the distinction from algorithm aversion, reliance on AI advice can be quantified separately using *Weight on Advice* (WOA), defined as

$$WOA = \frac{Revised\ Estimate - Initial\ Estimate}{AI\ Suggestion - Initial\ Estimate}.$$

Higher WOA values indicate greater reliance on AI advice, whereas lower values indicate stronger behavioral aversion to the advice [112]. We therefore treat WOA as a behavioral outcome of advice use, while treating the IAT score as the measure of the latent construct of implicit prejudice against machine [112].

C Appendix: Fixation and Saccade Features

Table 1: Fixation and Saccade Features

Dimension	Feature	Description	Ref
Fixation Features	**Current fix duration**	Total duration in 10 milliseconds during which the eye remains relatively stationary. Longer durations indicate deeper visual information processing.	[65]
	Current fix pupil	Pupil diameter during fixation. Changes in pupil size reflect cognitive load, emotional state, or visual attention.	[50]
	Current fix blink around	Blink events occurring around the fixation period. Blink patterns assess attention, cognitive workload, and potential distractions or fatigue.	[47]
	Current fix x resolution	The horizontal pixel-per-degree resolution for the current fixation.	-
	Current fix y resolution	The vertical pixel-per-degree resolution for the current fixation.	-
Saccade Features	**Current sac duration**	Time interval in 10 milliseconds during which a saccadic eye movement occurs. Longer durations suggest more complex movements.	[125]
	Current sac amplitude	Angular distance in degrees between the start and end points of a saccade. This reflects the extent of eye movement.	[117]
	Current sac avg velocity	Average velocity in degrees per second (°/s) during a saccade. Represents the speed of eye movement.	[30]
	Current sac direction	Angular direction of the saccadic movement, indicating the spatial direction of the gaze shift.	[127]
	Current sac peak velocity	Maximum velocity reached during a saccade. Indicates the intensity or forcefulness of the movement.	[96]

D Appendix: Model Architectures Summary

Table 2: Model Architectures

Model Type	Saccade Branch and Fixation Branch	Final Processing	Advantages	References
CNN-Fusion	Standard 1D CNN layers with ReLU and MaxPooling	Concatenation + Linear layers + ReLU + Dropout	Effectively captures modality-specific temporal patterns, enhancing feature complementarity	[87, 131]
MultiScale CNN	Dual-branch depthwise separable CNNs with channel attention per branch	Feature concatenation + Linear + GELU + Dropout	Extracts multi-scale temporal features and assigns relevance weights, improving sensitivity to nuanced temporal dynamics	[6, 85]
TransFusion	CNN projection + Transformer encoders (2 layers for sac, 3 for fix)	Mean pooling + Cross-modal attention + MLP	Models long-range dependencies and cross-dimensional interactions in time series, enabling richer representation learning for implicit cognitive states	[86, 122]

E Appendix: Preprocessing of Eye-Tracking Time-Series Data

This study developed a multi-stage preprocessing protocol to construct standardized eye-movement time-series datasets. For categorical features including saccade direction (`CURRENT SAC DIRECTION`) and fixation blink status (`CURRENT FIX BLINK AROUND`), label encoding

Table 3: Cross-fold held-out test performance with 95% confidence intervals.

Model	Accuracy	F1	AUC	Balanced Accuracy
Linear (baseline)	0.701 [0.694, 0.707]	0.822 [0.813, 0.832]	0.546 [0.508, 0.583]	0.506 [0.489, 0.522]
CNN-Fusion	0.705 [0.699, 0.710]	0.825 [0.819, 0.831]	0.575 [0.509, 0.640]	0.511 [0.492, 0.529]
MultiScale CNN	0.735 [0.708, 0.761]	0.831 [0.821, 0.842]	0.718 [0.652, 0.784]	0.601 [0.527, 0.676]
TransFusion	0.703 [0.703, 0.703]	0.826 [0.826, 0.826]	0.502 [0.488, 0.517]	0.500 [0.500, 0.500]

Note. Values are means with 95% confidence intervals in brackets, computed across the five held-out test folds.

was applied, with an "Unknown" category introduced to improve robustness to unseen labels. Numerical missing values were zero-filled, while categorical missing values were assigned default identifiers to ensure complete feature matrices. Temporal sequence alignment was performed by truncating or padding the data to 133 steps for saccades and 291 steps for fixations, based on the 95th percentile of the temporal length distributions. We then applied independent MinMax normalization to the saccade and fixation modalities to preserve within-modality feature relationships [4]. To improve robustness, we further adopted modality-specific augmentation, including temporal warping (maximum offset ±20%) for saccade features and Gaussian noise ($\sigma = 0.05$) for fixation features [115]. The final standardized dataset consisted of temporally aligned saccade matrices (5×133), fixation matrices (5×291), and binary prejudice labels, organized as 3D tensors (sample × feature × temporal sequence) for deep learning.

F Appendix: Model comparison method and result od Module 2

We selected four prominent models, including DeepSeek-R1 [28], GPT-4o [89], Claude-3-5-sonnet [8], and Grok-3 [128], as testbeds to investigate their varying approaches to developing adaptive interaction interfaces based on predicted prejudice results. Specifically, for each LLM, we conducted 120 agent-based experiments (60 per scenario) — one in which users exhibited implicit prejudice against the system and one in which they did not — with the aim of enhancing user experience by tailoring responses to individual user characteristics. Interaction strategies were organized along three dimensions: Involvement, Aesthetics, and Affect (Turner, 2017). In recent years, personalized design and deployment in human-AI collaboration contexts have received widespread attention [17, 20, 24], making user-specific strategy adaptation a key design consideration [77].

For analysis, we first coded the models' outputs and employed the Chi-squared (χ^2) test [108] to assess distributional differences among the four models, quantifying strategic variance attributable to their distinct architectures and training regimes. Next, we calculated and compared the distribution of strategy dimensions selected by each LLM across the 60 experiments per scenario. Finally, to enable deeper model-specific insights, we extracted the most frequently applied adaptive measures for further examination.

Our preliminary comparative analysis of the four LLMs revealed statistically significant differences in their strategic response patterns. The divergence was evident in both the prejudice condition ($\chi^2 = 32.95$, $p < .001$) and the non-prejudice condition ($\chi^2 = 136.49$, $p < .001$), as well as in the specific measures selected within each strategy. These differences, captured quantitatively via Chi-squared, underscore the existence of "model personalities"—distinctive behavioral profiles arising from variations in training data, alignment regimes, and architectural choices. GPT-4o prioritizes the Affect dimension, aligning closely with the Reinforcement Learning from Human Feedback (RLHF) strategy that optimizes perceived helpfulness and emotional resonance [91]. In contrast, Claude 3.5 Sonnet and DeepSeek-R1 excel in Aesthetics, though through different mechanisms. Claude 3.5 Sonnet's advantage likely reflects the principled coherence of its Constitutional AI framework [10], whereas DeepSeek's aesthetic edge may stem from large-scale, domain-specific training—such as code—that fosters mastery of intricate stylistic and structural patterns. These findings align with HCI research on LLM behavioral leanings [133], confirming that leading LLMs differ in their strategic outputs as a direct consequence of foundational design and sociotechnical influences. For the subsequent in-depth analysis in the main text, DeepSeek was selected as the primary testbed due to its marked and clear strategic divergence across the two scenarios.

G Appendix: Model Validity and Bounded Use Analysis

To further clarify the validity and bounded use of the Module 1 prediction model, we report additional uncertainty and error analyses. These supplementary analyses include fold-level confidence intervals, the confusion matrix of the best-performing MultiScale CNN fold, and false-positive and false-negative patterns. The purpose of these analyses is not to claim that the current model is a production-ready universal detector, but to characterize the current model's reliability and limitations as a proof-of-concept component for low-risk adaptive interface behavior. As shown in Table 3, the MultiScale CNN achieved the strongest cross-fold performance among the evaluated models, with a mean test accuracy of 0.735, a mean F1 score of 0.831, and a mean AUC of 0.718. These results are consistent with the main model comparison, but they also show non-trivial fold-level uncertainty. Thus, the model provides useful evidence that gaze features contain predictive information about implicit prejudice against machines, while still requiring larger datasets and stronger validation before broad deployment.

Tables 4 and 5 show that the best-performing MultiScale CNN fold had high sensitivity but limited specificity. In other words, the model was more likely to over-trigger prejudice-aware support than to miss prejudice-labeled samples. This error profile is important for bounding the model's appropriate use. The current model should not be used to assign stable psychological labels to users, make high-stakes

Table 4: Confusion matrix of the best-performing MultiScale CNN fold.

	Predicted non-prejudice	Predicted prejudice
Actual non-prejudice	57	95
Actual prejudice	29	331

Note. The positive class is implicit prejudice against machine. The matrix is based on the held-out test set of the best-performing MultiScale CNN fold.

Table 5: False-positive and false-negative analysis for the best-performing MultiScale CNN fold.

Metric	Value
False positives	95
False negatives	29
Sensitivity / Recall	0.919
Specificity	0.375
False-positive rate	0.625
False-negative rate	0.081
Precision / Positive predictive value	0.777
Negative predictive value	0.663
Balanced accuracy	0.647

decisions, or serve as a universal detector of implicit prejudice. Instead, it should be understood as a proof-of-concept component that can inform low-risk, reversible interface adaptations, such as providing additional explanation, transparency, affective support, or user control. This interpretation is especially important because the current results are based on a modest dataset. Although the results suggest that non-intrusive gaze signals can support prejudice-aware adaptation, they do not yet establish generalization or cross-task robustness. Future work should therefore validate the model on larger samples, unseen participants, and additional human-AI collaboration tasks before using it in more general or consequential deployment settings.

H Appendix: SHAP-Based Model Interpretation

SHAP (SHapley Additive exPlanations), a model interpretation method grounded in the Shapley Value from game theory [81], quantifies the contribution of each feature to model outputs. In this study, we first select the best-performing model for each prediction target as the subject of explanation. All samples from the test set are used as the background data, and their temporal lengths are aligned to a unified number of time steps. Subsequently, a wrapper class is defined to convert the core model from a multi-input structure into a single-input format, enabling compatibility with the SHAP explainer. The `shap.GradientExplainer` is then employed by providing the wrapped model and background data as inputs, which leverages model gradients to compute the contribution of each input feature.

The Shapley value for feature i is defined as:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} \left[f(S \cup \{i\}) - f(S) \right] \tag{1}$$

Consider a complete feature set N with cardinality $|N| = M$. To evaluate the individual contribution of a feature i, we examine model predictions $f(S)$ based on subsets $S \subseteq N \setminus \{i\}$, which omit feature i. Here, $f(S)$ represents the model's output when only the features in S are provided, interpreted as an expected value. The corresponding Shapley value ϕ_i quantifies how much feature i contributes to the model's prediction.

In practice, for each test instance, SHAP values are divided into two groups: the top five channels capture saccade-related features, while the remaining ones pertain to fixation-related features. To preserve the directionality of feature influence, we compute the mean Shapley value across the entire sample set as follows:

$$\text{MeanContribution}(x_i) = \frac{1}{N} \sum_{j=1}^{N} \phi_i^{(j)} \tag{2}$$

where N is the number of samples and $\phi_i^{(j)}$ denotes the SHAP value of feature x_i in the j-th sample. The absolute value operator $|\cdot|$, when used, captures the magnitude of impact without regard to directionality.

I Appendix: Full model performance comparison of Module 1

Table 6: Full model performance comparison on the validation and test sets.

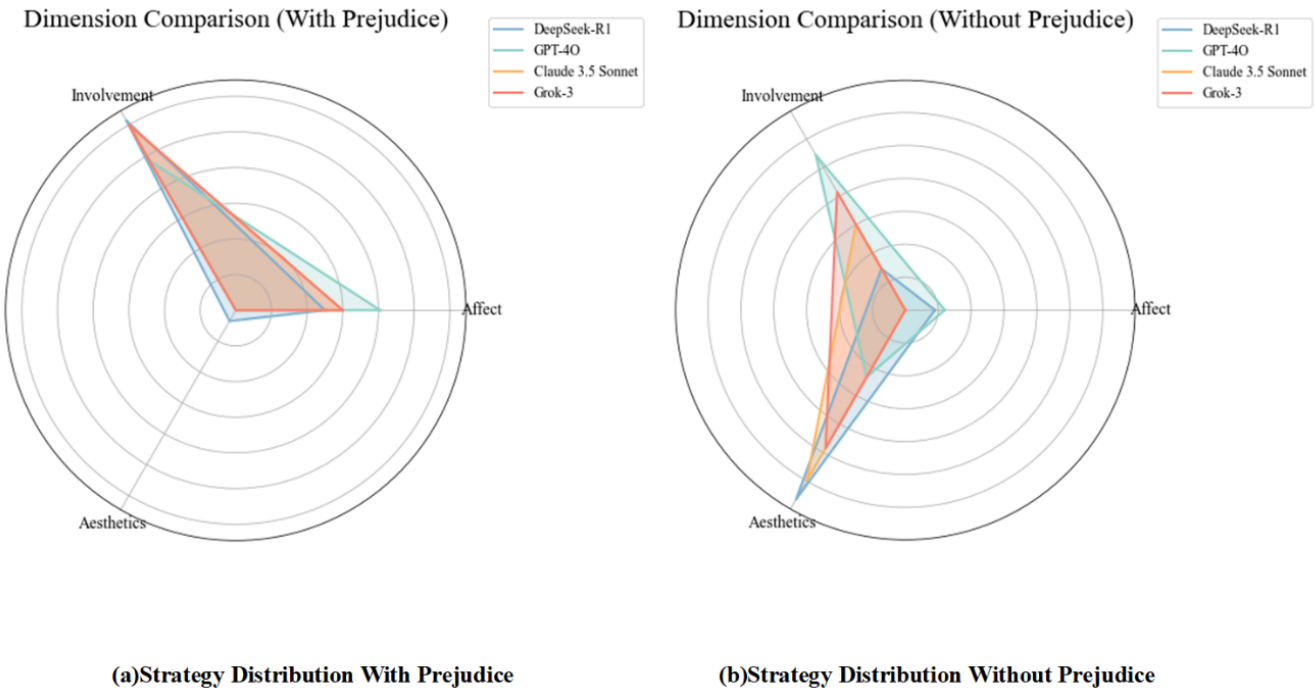
Model	Type	Accuracy	Precision	Recall	F1	AUC	Hamming Loss
Linear (baseline)	Val	0.704	0.704	1.000	0.826	0.510	0.296
	Test	0.703	0.703	1.000	0.826	0.505	0.297
CNN-Fusion	Val	0.710	0.709	0.997	0.828	0.603	0.290
	Test	0.713	0.712	0.994	0.830	0.624	0.287
MultiScale CNN	**Val**	**0.773**	**0.789**	**0.924**	**0.851**	**0.757**	**0.227**
	Test	**0.758**	**0.777**	**0.919**	**0.842**	**0.753**	**0.242**
TransFusion	Val	0.702	0.702	1.000	0.825	0.486	0.298
	Test	0.703	0.703	1.000	0.826	0.507	0.297

J Appendix: Expert Interview participant Data

Table 7: Participant Information and Responses

ID	Gender	Age	Research Field	Education	Biased	Unbiased	Key Opinion
1	Female	21	Applied Statistics	Master's	5	5	User-centered design.
2	Male	24	Master of Finance (Corporate Finance)	Master's Student	5	6	Affect dimension is the most important.
3	Male	23	Anomaly Detection in Home Environments	PhD Student	6	5	Enhance the experience and reduce the sense of foreignness.
4	Female	23	Research on Analogous Theories in the Field of Complex Systems	Master's	5	6	When users exhibit prejudice, it's important to leverage the core functions of AI to eliminate that prejudice — that is, to optimize involvement and make AI more useful and reliable. In this case, I believe aesthetics should be a lower priority. When users are without prejudice, it indicates a high level of acceptance and expectation toward AI. At this point, we should expand AI's feasible boundaries, optimize its appearance, enhance user experience, and strengthen user reliance.
5	Male	23	Human Factors in Language-Based Warning Interaction	Master's	4	5	Improve the core user experience, emphasize strengths, and hide elements users dislike.
6	Male	22	Swarm Control of Unmanned Systems	Master's	6	5	User experience counts.

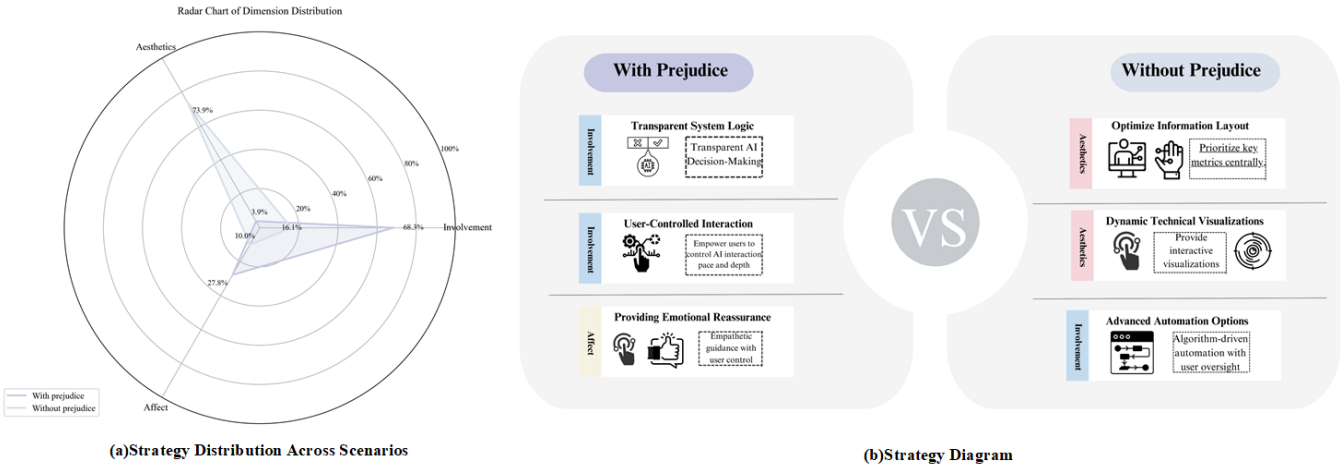
K Appendix: Module 2 Strategic Response Patterns Across Prejudice Conditions



(a)Strategy Distribution With Prejudice **(b)Strategy Distribution Without Prejudice**

Figure 2: Strategic Response Patterns Across Prejudice Conditions

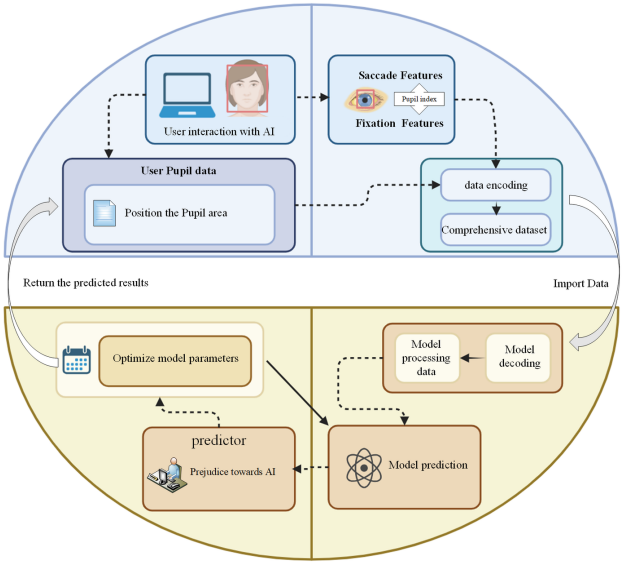
L Appendix: Module 2 LLM-Generated Strategy Patterns Across Prejudice Conditions



(a)Strategy Distribution Across Scenarios **(b)Strategy Diagram**

Figure 3: Strategy Distribution and Diagram

M Appendix: Module 3 System Design



Algorithm 1 Human-AI Collaboration System for Predicting implicit prejudice against machine

Input: User interaction data (text + eye movement features)
Output: implicit prejudice against machine

1: **call** InitSystem ▷ Initialize Vue frontend, Flask backend, and establish Socket connection
2: **EYE_FEATURES** ← CaptureEyeFeatures() ▷ Extract saccade, fixation, and pupil index features during interaction
3: **ENCODED_DATA** ← EncodeData(EYE_FEATURES) ▷ Merge and encode into a comprehensive dataset
4: ImportDataset(ENCODED_DATA) ▷ Send dataset to backend for prediction
5: ImportModel() ▷ Load pretrained bias prediction model
6: **PROCESSED_DATA** ← ModelDecoding(ENCODED_DATA) ▷ Transform input data to model-ready format
7: **RESULT** ← ModelPrediction(PROCESSED_DATA) ▷ Predict implicit prejudice against machine using encoded data
8: **call** Predictor(RESULT) ▷ Interpret output as a prejudice indicator
9: OptimizeModelParameters(RESULT) ▷ Perform parameter tuning for improved accuracy
10: ReturnToFrontend(RESULT) ▷ Send implicit prejudice result back to the Vue UI
11: **return** "implicit prejudice completed"

Figure 4: System Design

N Appendix: Module 3 Implementation-Level System Architecture

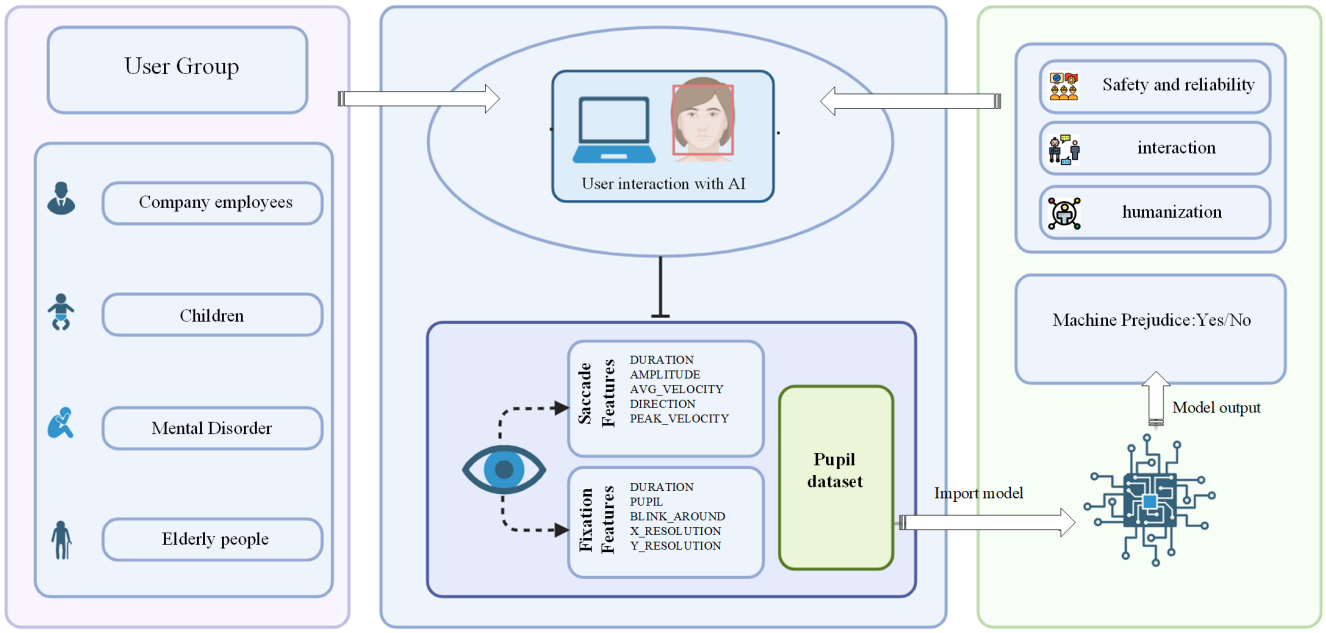


Figure 5: System Architecture

O Appendix: Module 3 architecture details

In the deployed system, webcam-based gaze estimation is handled primarily by WebGazer, which outputs screen-space gaze coordinates and timestamps for each frame in the browser [57, 71]. Any face- and eye-region localization used at runtime is treated as an auxiliary step

Table 9: Full Holm-corrected pairwise comparisons based on participant-level scenario means.

Indicator	Comparison	Δ	90% CI	d_z	p_t	p_{Holm}
Cognitive trust	Random Strategy – No Strategy	0.247	[-0.062, 0.556]	0.34	.182	.182
	PrejSense – Random Strategy	0.235	[0.016, 0.454]	0.46	.078†	.157
	PrejSense – No Strategy	0.482	[0.161, 0.803]	0.64	.018*	.055†
Emotional trust	Random Strategy – No Strategy	0.353	[0.094, 0.612]	0.58	.030*	.030*
	PrejSense – Random Strategy	0.565	[0.284, 0.846]	0.85	.003**	.007**
	PrejSense – No Strategy	0.918	[0.473, 1.363]	0.87	.002**	.007**
WOA	Random Strategy – No Strategy	0.048	[-0.036, 0.132]	0.24	.334	.334
	PrejSense – Random Strategy	0.104	[0.013, 0.195]	0.48	.064†	.127
	PrejSense – No Strategy	0.152	[0.047, 0.257]	0.61	.023*	.068†

Note. Δ is calculated as the latter scenario minus the former scenario. The 90% confidence intervals are reported to align with the $p < .10$ marginal-effect threshold used for this small-sample proof-of-concept user test. d_z denotes the standardized mean difference for paired comparisons. p_t reports the two-sided paired-sample t test. p_{Holm} reports Holm-corrected p values within each indicator across the three pairwise comparisons. †$p < .10$, * $p < .05$, ** $p < .01$.

for eye-region alignment, blink detection, and pupil-related measurements, rather than as a second independent gaze-estimation pipeline. After acquisition, the gaze and pupil streams are preprocessed to reduce noise and tracking artifacts, including smoothing, removal of blink-related or invalid segments, and normalization when needed [52, 54]. The cleaned sequences are then segmented into fixations and saccades using standard event-detection rules [99], and the system derives the same fixation- and saccade-related variables used in Module 1, including current fix duration, current fix pupil, current fix blink around, current sac amplitude, and current sac avg velocity. For online operation, these event streams can be updated continuously with an online eye-movement classifier [37], while blink-related cues can be refined by a pretrained vision model when available [101]. The extracted features are formatted to match the pretrained Module 1 input representation and are then forwarded to the prejudice predictor, whose output is used by Module 2 to select the corresponding adaptive strategy. This formulation preserves a clear separation between sensing, prediction, and adaptation, and keeps Module 3 consistent with the overall three-module architecture of PrejSense.

P Appendix: Indicators' details of User Testing

Table 8: Indicators of User Testing

Variable	Definition	Reference
Implicit Prejudice	Social preferences outside conscious control.	[112]
Algorithm Aversion	Users' unwillingness to accept and utilize advice from algorithms, even though such algorithms can be more accurate than humans yet probabilistically imperfect.	[112]
Emotional Trust	Trust formed through emotional, intuitive responses to AI, driven by feelings, empathy, and human-like cues rather than rational analysis.	[21]
Cognitive Trust	Trust based on rational evaluation of AI's reliability, competence, and transparency using objective evidence and consistent performance cues.	[21]